

Response to the reviewers

Manuscript: *Minimum Weighted Hazard-Exposure Dispatch: Complexity, an Exact Algorithm, and an FPTAS*

Journal of Combinatorial Optimization — major revision

Dear Editor and Reviewers,

Thank you for the careful reading and the constructive reports. We revised the manuscript substantially. Reviewer 1’s central objection, that the theoretical results follow from the classical theory of $1\|\sum w_j U_j$, is correct, and we have rewritten the abstract, introduction, related work and summary table to say so plainly, to cite the classical sources for each result, and to separate what is classical from what we believe is new. Reviewer 1’s question about the FPTAS exposed a real weakness in our experiments (the FPTAS’s rounding was inactive in most of them), which we explain and fix. Reviewer 2’s requests for a clearer hardness proof, a self-contained proof of Theorem 5, a shorter positioning section and a discussion of parameter uncertainty are all addressed.

While revising we also found and corrected several errors of our own, which we list in Section 3. Theorem numbers 2–5 and Proposition 6 are unchanged; new results are Proposition 7, Proposition 8 and Theorem 9, and a new “Fact 1” used in the proof of Theorem 2. All numerical tables were regenerated.

Summary of the main changes

1. **Positioning and novelty** (abstract, Sections 1, 2, 4, Table 2, conclusion): the results that are classical are labelled as such, with the classical references added (Sahni 1976; Gens and Levner 1981; the Lawler–Moore dynamic program is identified with Theorem 3).
2. **New or sharpened results** (Section 4): a classification of MWHED restrictions by which of p, w, d are constant (Theorem 9); Theorem 5 now covers m identical vehicles and has a complete, self-contained proof; unbounded worst-case ratio of deadline-order dispatch with and without skipping (Proposition 6) and of the weighted greedy repair (Proposition 7); tightness of the FPTAS analysis up to a factor $1 - \epsilon^2$ (Proposition 8).
3. **Hardness proof** (Theorem 2): rewritten as a direct reduction from PARTITION with an explicit lemma (Fact 1) on feasibility under a common deadline.
4. **Experiments** (Section 5): redesigned, with exhaustive-search verification, instance sizes to $n = 400$, a stronger baseline, FPTAS experiments in which rounding is active, and an adversarial family.
5. **Case study** (Section 5.6): deadline semantics clarified (Remark 1), results recomputed, sensitivity analysis added.
6. **Discussion** (Section 6): the multi-vehicle subsection is corrected, and parameter uncertainty is treated in a new subsection with the two suggested references.

1 Reviewer 1

While the manuscript is technically correct and generally well presented, the level of novelty of the theoretical contributions is not sufficiently convincing. [...] most of the main theoretical

results appear to follow from well-known results in the scheduling literature. [...] Consequently, the manuscript does not clearly establish new complexity results, new algorithmic techniques, or substantially new structural properties beyond those already known for the equivalent scheduling formulation.

Response. We agree. The manuscript proves that MWHED is $1 \parallel \sum w_j U_j$ and the general results follow from it. We have not tried to argue otherwise. Concretely:

- The abstract and the introduction (Section 1) now say that the basic complexity picture is inherited from scheduling theory and that we claim no new technique for the general problem. The contributions are split into two lists: a self-contained development of the classical results in transportation language (Theorems 2–4), and what we believe the framing adds (Theorems 5 and 9, Propositions 6–8).
- Section 2 now states, for each of Theorems 2–5, which classical result it is: Theorem 3 is the Lawler–Moore dynamic program, Theorem 4 is the classical value-scaling scheme for job sequencing with deadlines (we added Sahni 1976 and Gens and Levner 1981, which we had missed), Theorem 5 for one vehicle is classical unit-time job sequencing with profits, and the hardness argument is the knapsack equivalence. The sentence in the earlier Section 2 suggesting that the FPTAS adaptation was “not entirely mechanical” has been removed, since that overstated the point.
- Table 2 (the summary table) has a new column giving the origin of each result (classical / Lawler–Moore / Moore / new).
- We removed the claim in the abstract, introduction and conclusion that the dynamic program and the FPTAS are “necessary” for any guarantee. Proposition 6 shows only that deadline-order dispatch has no guarantee; it does not show that other polynomial algorithms with guarantees cannot exist, and the text now says so.

1. Could the authors identify a new structural property?

Response. We identify the following, with the caveat that each is elementary once the equivalence is in place; we present them as modest additions, not as deep results.

- **Theorem 9 (classification by homogeneity).** For every restriction obtained by making some of the dispatch times p , weights w and deadlines d constant, MWHED is polynomial if p or w is constant and weakly NP-hard otherwise (already when only d is constant). In particular heterogeneous hazard deadlines never cause hardness. This turns the informal question of which network structure makes dispatch hard into a precise statement, and it is consistent with the parameterized results of Heeger and Hermelin (2024), which we now discuss accurately (see Section 3).
- **Theorem 5 for m identical vehicles.** The equal-cost matroid greedy extends to any number of identical vehicles, with a complete proof (slots at each position, a counting criterion for feasibility, the augmentation property, optimality of the greedy, and a union–find implementation whose correctness we prove). The submitted discussion suggested that the equal-cost case would become hard with several vehicles; that inference was wrong (see Section 3). Release dates, which the cited hardness result of Heeger and Molter (2025) needs, are what break tractability.

- **Propositions 6–8.** Deadline-order dispatch with and without skipping, and the weighted greedy repair, have unbounded worst-case ratio, and the standard FPTAS analysis is tight up to a factor $1 - \epsilon^2$ (the adversarial family also explains empirically why the FPTAS is nearly exact on random instances, see below).

2. The manuscript repeatedly states the dynamic program and FPTAS as major contributions. However, the relationship between these algorithms and classical Lawler-Moore-type methods is not discussed in sufficient detail. Are the algorithmic contributions truly different from known scheduling algorithms?

Response. No. Theorem 3’s dynamic program is the Lawler–Moore dynamic program (jobs in due-date order, state equal to the accepted processing time, one test per accepted job); Remark 2 (after Theorem 3) says so and describes exactly what our proof adds (a derivation from the earliest-deadline-first lemma in the notation of the dispatch problem). Theorem 4 is the classical value-scaling scheme, which Sahni (1976) and Gens and Levner (1981) developed for job sequencing with deadlines; Section 2 (“Scheduling with deadlines” and “Approximation”) now states this, notes that faster schemes are known, and explains why we keep the classical construction (its analysis is the cleanest, and Proposition 8 shows it is tight). We no longer present either algorithm as a major contribution; they are presented as a self-contained treatment for a transportation audience.

3. Why does FPTAS = 1000 always appears in every table? Could the authors strengthen the empirical evaluation?

Response. This was a genuine weakness. The reason is that, with weights at most 100, the scaling factor $K = \max(1, \epsilon w_{\max}/n)$ equals 1 in almost all of our runs, so no rounding took place and the “FPTAS” coincided with the exact value-indexed dynamic program. A ratio of 1.000 was therefore guaranteed and said nothing about the approximation scheme. We now state this explicitly (Section 5.2, last row of Table 4: the share of instances with $K = 1$) and have redesigned the study (Section 5; every table was regenerated):

- **Verification** (Section 5.1): every algorithm is checked against exhaustive search, including all $n!$ dispatch orders for $n \leq 7$ (400 instances), best-feasible-subset enumeration for $n \leq 14$ (1,500 instances, one third with weights up to 10^6 , with the FPTAS guarantee checked in 4,500 runs of which 2,923 had active scaling), and exhaustive search over vehicle assignments for the equal-cost algorithm with $m = 2, 3$. There were no violations.
- **Accuracy across sizes** (Section 5.2): n up to 400, 50 instances per size (previously ≤ 100 and 20), worst-case ratios reported as well as means, and a stronger baseline, EDD with skipping, in addition to naive EDD.
- **FPTAS with active scaling** (Section 5.3, Tables 5–6, Figure 3): weights up to 10^6 (so K is in the hundreds to thousands), a strongly correlated family, and the adversarial family of Proposition 8. Findings: on random and strongly correlated instances the FPTAS is nearly exact (worst ratio 0.9993 even at $\epsilon = 0.5$) because the bound $\epsilon w_{\max}$ is pessimistic when the optimum contains many sites; on the adversarial family the ratio falls to 0.669 at $\epsilon = 0.5$ and follows the guarantee as ϵ shrinks, so the guarantee is not vacuous. A simple completion step (re-insert rejected sites greedily) repairs the family completely.

- **Runtime** (Section 5.4): re-run with active scaling.

We chose not to report this as “FPTAS = 1” evidence at all, and the text says explicitly that the earlier tables should not be read as evidence about the scheme.

2 Reviewer 2

3. *The part of MWHEd’s positioning in page 6 is comprehensible, nevertheless it could be rephrased better by shorter arguments.*

Response. The “Positioning” paragraph (Section 2) has been cut by roughly half; it now lists the four literatures it draws on and refers to Table 2 for which results are classical.

1. *Theorem 2 on NP-hardness of MWHEd might require a bit efforts to get full clarity. In the 3rd line from the bottom of page 9, don’t you think that the fact “completion times along a fixed set are additive” is too strong? What fixed sets should we use here? [...] The authors should explain about additivity of special sets [...] since it is not obvious by default, feasibility condition does not explicitly implies additivity. [...] the above condition ($\diamond$) $\sum_{i \in S} p_i \leq D$ is crucial to relate your formulation to a 0/1 knapsack instance.*

Response. Thank you; the sentence was imprecise. We replaced it by an explicit statement, **Fact 1** (Section 4.1), with a proof: *if all deadlines equal D , a set S of sites can all be served on time iff $\sum_{i \in S} p_i \leq D$* , which is exactly your condition ($\diamond$). The “fixed set” is an arbitrary subset S . The only additivity used is that the *last* site served from S completes at the sum of all of S ’s occupancies, because the vehicle is never idle and each p_i does not depend on the predecessor (the defining property of MWHEd); this gives necessity. For sufficiency every completion time is a partial sum of positive numbers and so is at most the total. The proof of Theorem 2 now cites Fact 1, and the knapsack correspondence for general weights is stated after the theorem.

2. *[...] your claim on “NP-complete of the 0/1 knapsack problem” comes from polynomial reduction of PARTITION (one of the six popular graph-based NP-complete problems [...]). This is a solidly smart argument, if the above concern is clearly elucidated. Otherwise, a graph-based pure proof for Theorem 2 possibly might give better insights (ref. Delazeri and Ritt (2026)).*

Response. We removed the dependence on the knapsack citation: Theorem 2 now gives a *direct* polynomial reduction from PARTITION to MWHEd ($p_i = w_i = a_i$, $d_i = A/2$), so that the proof is self-contained and also shows hardness when $w_i = p_i$; membership in NP is added. A small remark on terminology: PARTITION is one of the six basic NP-complete problems in Garey and Johnson but it is a number problem, not a graph problem (the graph-based members of that list are Vertex Cover, Clique and Hamiltonian Circuit). We did not give a graph-based proof: MWHEd has no graph structure once each visit is a round trip from the depot, so there is no natural graph on which to base one. Delazeri and Ritt (2026) prove NP-completeness for a different, graph-based model of wildfire suppression with simultaneous resources; Section 2 says so.

3. *[...] the authors would elucidate more explicitly Theorem 5’s proof. In particular, would you remind readers the concepts (a) transversal matroid, union-find structure, and (b) the related notation $\alpha(n)$?*

Response. Section 4.4 now opens with a paragraph recalling the notions used: matroid and its three axioms, the greedy algorithm, the transversal matroid (sites against (vehicle, position) slots), and the union–find structure with path compression, with α defined as the inverse Ackermann function. The proof is rewritten in five steps: (1) positions and slots; (2) a counting criterion for feasibility, $N_S(k) \leq mk$ for all k ; (3) a direct proof that the feasible sets satisfy the matroid augmentation property (so the transversal-matroid fact is proved, not cited); (4) a short proof that the greedy algorithm is optimal on any matroid; (5) the union–find implementation, with a proof that it accepts a site exactly when the kept set plus that site is feasible. Algorithm 3 is rewritten accordingly, and Example 5 now also shows $m = 2$ vehicles.

4. [...] do we agree that the problem of MWHED is obviously data-driven, the proofs and optimal solutions obtained depend on both accurate formulation and precise calibration of model parameters, upon uncertainty? To some extent, would formulation of uncertainty [...] (at not just variable level but also at parameter one) be considered in next studies, where stochastic programming might be a potential approach? Kindly, e.g. refer to problems of evacuation planning under unsafe contexts and facility location under disruptive events (FLDE), ref. [1, 2].

Response. We agree. Section 6.3 (now “Uncertain parameters”) was rewritten to cover uncertainty in p_i , d_i and w_i , not only in the deadlines. It discusses (i) scenario-based and two-stage stochastic formulations and which of our results survive (the earliest-deadline lemma holds scenario by scenario but not simultaneously; a worked example shows that sorting by expected deadline can be nearly twice as bad as optimal), citing your two references (Wang, 2020; Koca, 2023) (we verified both: Computers & Industrial Engineering 145, 106458, and 183, 109484; the database lists a single author for the second); (ii) a robust-feasibility observation (feasibility is monotone in the parameters, so a set that is on time for every realization in a box is a feasible set of one MWHED instance, which also shows that full robustness is too conservative in our case study); and (iii) calibration and sensitivity analysis. In addition, the case study (Section 5.6) now includes a Monte-Carlo sensitivity analysis over response speed, arrival times and departure delay, which shows that the nominal-optimal plan is brittle (fully on time in 20.7% of 5,000 draws) while the simpler plan “serve Paradise only” is robust. Stochastic programming for MWHED itself is listed as the main direction for future work; we have not attempted a solution method for it here.

3 Errors of our own corrected in this revision

In the course of revising we re-checked the manuscript and the code and found the following, which we correct and disclose.

1. **Case study: deadline semantics.** The model’s on-time test counts the completion of the whole round trip (the “return reading”), but the narrative and the abstract speak of reaching the site before the hazard (the “arrival reading”). The submitted case study used the return reading while describing arrival, and consequently claimed that Concow and Magalia could not be reached at 50 km/h although the crew arrives after 31.5 and 44.5 minutes against fire-arrival minutes 52 and 58. Remark 1 now makes the two readings precise (the arrival reading is an instance of MWHED with deadlines $d_i^{\text{haz}} + p_i/2$) and the case study uses it. The results change: at 80 km/h the optimum is Concow then Paradise (26,928), not Paradise alone. The return reading is reported for comparison.

2. **Case study: baselines and wording.** Individually infeasible sites are now deleted before every algorithm, including the naive baseline (the submitted “naive = 0” result used sites that violate Assumption 1, so it did not illustrate Proposition 6). Statements that the optimum “saves” a given number of people were replaced by the on-time weight (population) of the sites served.
3. **Naive baseline.** Our naive EDD baseline kept dispatching to sites that were already late, which made it a weak anchor; we added EDD with skipping.
4. **Heeger–Hermelin.** The submitted text said that a small number of distinct dispatch costs “does not guarantee tractability”. The result is $W[1]$ -hardness in that parameter (no $f(k)\text{poly}(n)$ algorithm), while for fixed k the problem is polynomial; the text now says this.
5. **Multiple vehicles.** The claim that the $m = 1$ instance “embeds” by adding idle vehicles was wrong (extra vehicles can serve sites); we now use blocker sites. The inference from Heeger and Molter (2025) that the equal-cost case becomes hard with several vehicles was also wrong, because that result requires release dates, which MWHED does not have; Theorem 5 now shows tractability for every m .
6. **Implementation.** The FPTAS guarantee relies on Assumption 1 (so that $w_{\max} \leq W^*$), which the code did not enforce; both algorithms now delete infeasible sites first, and the code was rewritten and checked against exhaustive search.
7. Minor: the dash convention in the dynamic-program example table was described incorrectly; the explanation of why the FPTAS runtime was flat in ϵ cited saturating weights, whereas the cause was $K = 1$. Both are fixed.

We believe the revised manuscript is considerably stronger and more honest about what is and is not new, and we thank the reviewers for the comments that led to these changes.

Sincerely,
The authors

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